Mathematical Background: Kernels Phil 233

Preliminary: From ODEs to PDEs

Wilson briefly argues that we might be misled by celestial mechanics: Newtonian N-body dynamics is an anomalously *simple* setting; conceptual machinery of applied mathematics is not needed. The contrast:

ODE setting (point-mass mechanics):

- State = finitely many numbers (positions, velocities of N particles); evolution governed by $3N$ coupled ODEs (derived from $F = ma$ with specified forces).
- **Initial conditions** determine the solution: data at one instant t_0. **Boundary conditions**: the solar system is treated as an "island universe," specified number of interacting bodies (rule out "space invaders"). (See Earman *Primer on Determinism* and Wilson's review, subsequent debates on fate of determinism – Norton's dome and responses.)

PDE setting (continuous bodies, fields):

- State = a *function* (a field defined over a spatial region); infinitely many degrees of freedom.
- Need initial conditions (data on a time slice), **boundary conditions** (data on surfaces extending through time), and possibly **interfacial conditions** where regions of differing composition meet.
- The **interior equation** and the **boundary/interface data** register information differently (one smears/averages over fine detail; the other condenses data onto lower-dimensional surfaces) and must "*harmonize*" (Wilson's term).
- New sources of complexity absent in the ODE case: spatial **scales** and cutoffs, **constraints**, **Fourier/eigenfunction decompositions**, questions of which function space the solution lives in, and existence/uniqueness/ well-posedness that can fail in subtle ways.

1. The Loaded-String Equation

String of length L, pinned at both ends, deflection $y(x)$ under downward load density $f(x)$:

$$-k\,\frac{d^2y}{dx^2} = f(x), \qquad y(0) = y(L) = 0. \tag{1}$$

- Interior equation: local curvature $\propto$ local load. (Sign convention: Wilson writes $k\,y'' = f$; the minus makes $-d^2/dx^2$ a *positive* operator.)
- Boundary conditions: string pinned at endpoints.
- Operator $L := -k\,d^2/dx^2$ is **linear**, BCs **homogeneous** $\Rightarrow$ responses to separate loads **superpose**. This is what allows us to use the integral representation.

2. Green's Function = Response to a Point Load

Solve the simplest load — a unit load concentrated at α, i.e. $\delta(x-\alpha)$ — then superpose. The response is the Green's function:

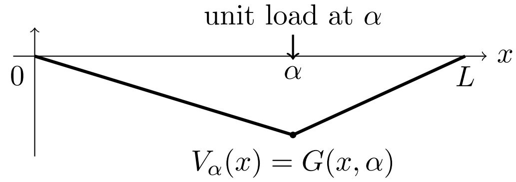
$$-k\,G_{xx}(x,\alpha) = \delta(x-\alpha), \qquad G(0,\alpha) = G(L,\alpha) = 0. \tag{2}$$

Derivation (schematic):

- Off the load point, $G_{xx} = 0 \Rightarrow G$ is linear on each side: $G = Ax$ for $x \le \alpha$, $G = B(L-x)$ for $x \ge \alpha$ (BCs already built in).
- Continuity at α: $A\alpha = B(L-\alpha)$.
- Slope-jump (integrate (2) across α): $A + B = 1/k$.
- Solve: $A = (L-\alpha)/(kL)$, $B = \alpha/(kL)$.

$$G(x,\alpha) = \frac{1}{kL}\begin{cases}(L-\alpha)\,x & 0 \le x \le \alpha,\\ \alpha\,(L-x) & \alpha \le x \le L.\end{cases} \tag{3}$$

This is Euler's $V_\alpha(x)$: a downward tent with a sharp peak under the load.



Two key properties:

- **Symmetry** $G(x,\alpha) = G(\alpha,x)$ (reciprocity) $\Leftrightarrow L$ is self-adjoint.
- **Corner at** $x = \alpha$: G_x jumps, G_{xx} undefined classically there. This is d'Alembert's worry, prompts the foundational worries (similar to stresses at corners of boxes and other issues Wilson discusses).

3. The Kernel and the Superposition Integral

Write a distributed load as a superposition of point loads, $f(x) = \int_0^L \delta(x-\alpha)f(\alpha)\,d\alpha$; superpose the responses:

$$y(x) = \int_0^L G(x,\alpha)\,f(\alpha)\,d\alpha. \tag{4}$$

- $G(x,\alpha)$ = the **kernel**: the two-variable function inside the integral specifying how the load at source point α contributes to the deflection at field point x.
- The integral operator $Tf = \int G(\cdot,\alpha)f(\alpha)d\alpha$ is the **inverse** of the differential operator: $T = L^{-1}$. (Apply L to (4), use (2), recover $-ky'' = f$.)
- Solving the differential equation $\rightsquigarrow$ evaluating an integral.

- The corner is **harmless inside the integral**: integrating the tents against a smooth f yields a genuinely twice-differentiable y. Hence the old practice of tolerating V_α only when "housed inside an integral as a kernel."

4. Why V_α Fails the *Classical* Test

- Classical "solution" = twice-differentiable y satisfying (1) pointwise everywhere.
- V_α has no second (or even first!) derivative at $\alpha \Rightarrow$ disqualified.
- Energetic gloss: a true corner stores infinite strain energy at a point — mechanically incoherent (d'Alembert vs. Euler).

5. The Weak (Distributional) Reading

Reinterpret "satisfying the equation." Multiply (1) by a smooth test function ϕ with $\phi(0) = \phi(L) = 0$, integrate by parts once:

$$\int_0^L f\,\phi\,dx = \int_0^L (-ky'')\phi\,dx = \int_0^L k\,y'\,\phi'\,dx. \tag{5}$$

- **Weak form** asks only for y', and only inside an integral against ϕ' — never y'' at a point.
- V_α has a fine first derivative a.e. (piecewise constant, one jump) $\Rightarrow$ it *satisfies* (5) with $f = \delta(\cdot - \alpha)$. It is a bona fide **weak solution**.
- Schwartz: redefine the derivative itself so the corner's distributional second derivative *is* $-\frac{1}{k}\delta(\cdot - \alpha)$; then $-kV_\alpha'' = \delta$ holds in the distributional sense.
- **Cooperative task allocation**: smooth ϕ's supply the derivative V_α lacks; V_α supplies the concentrated influence no smooth function can. Content is carried by the integrated system, not the interior formula in isolation — Wilson's reply to "single-sentence idolatry."

6. The General Pattern

- For a linear operator L with BCs: Green's function $L_x G(x, y) = \delta(x - y)$ is the kernel of L^{-1}; solution $u(x) = \int G(x, y) f(y)\,dy$.
- Constant coefficients on an unbounded domain $\Rightarrow G = G(x - y)$ (**convolution**), diagonalized by the Fourier transform (kernel $e^{-2\pi i x\xi}$).
- In every case: one two-variable function encodes how each source point influences each field point; the singularity at $x = y$ is exactly what makes the kernel invert a differential operator — harmless in the integral, ill-defined pointwise.
- Distribution theory puts the two-century-old kernel practice on rigorous footing — Wilson's case study in retroactive vindication.